Course Project 4
Non-Linearity, Tree-Based Classification, and SVM

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1 Motivation

This project studies a bank telemarketing problem using the Bank Marketing dataset of Moro et al. The practical objective is to improve future marketing campaigns by predicting whether a client will subscribe to a term deposit product. From a business perspective, the institution is not interested in calling every customer uniformly; rather, it seeks to identify clients with a higher probability of subscription so that campaign resources can be allocated more efficiently. The target variable is `deposit`, where `yes` indicates that the client subscribed to the product and `no` indicates that the client did not. This defines a binary classification problem on structured tabular data. The predictors describe four aspects of the campaign setting: customer profile (`age`, `job`, `marital`, `education`), financial status (`balance`, `default`, `housing`, `loan`), current campaign information (`contact`, `day`, `month`, `campaign`, `duration`), and previous campaign history (`previous`, `pdays`, `poutcome`).

Two variables require special attention. First, `duration` measures the last call length in seconds and is highly predictive, but it is only known after the call has occurred; therefore, it creates leakage and was excluded. Second, `pdays` contains the value `-1`, which is not a genuine numeric quantity but a code indicating no previous contact. This variable was re-expressed before modeling.

Our central analytical question is whether this prediction problem is adequately described by a linear decision rule, or whether more flexible methods — non-linear feature expansions, tree ensembles, and kernel SVMs — yield systematically better performance.

2 Data Analysis and Preprocessing

The dataset contains 11,162 observations and mixed data types. Initial exploratory analysis showed that the target distribution is relatively balanced, with approximately 52.6% `no` and 47.4% `yes`. Hence, accuracy is still meaningful, although it should not be interpreted alone.

Inspection of the numeric variables through histograms and boxplots revealed substantial skewness and outliers, especially in `balance`, `campaign`, `previous`, and `pdays`. In addition, the numeric correlation matrix did not show very strong linear associations overall, which suggested that purely linear models may be insufficient.

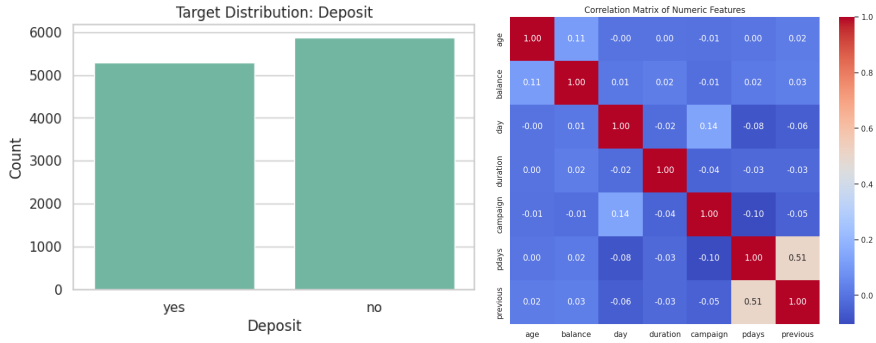

Figure 1: Exploratory analysis: class distribution, and numeric correlation structure.

The preprocessing pipeline was guided by these observations:

1. The target was converted to binary form:

$$y = \begin{cases} 1, & \text{if deposit} = \text{yes} \\ 0, & \text{if deposit} = \text{no} \end{cases}$$

2. `duration` was removed to avoid leakage.

3. `pdays` was decomposed into two variables:

$$\texttt{was_previously_contacted} = \mathbf{1}(\texttt{pdays} \neq -1)$$

   and

$$\texttt{pdays_clean} = \begin{cases} 0, & \text{if pdays} = -1 \\ \texttt{pdays}, & \text{otherwise} \end{cases}$$

4. Categorical variables were one-hot encoded.

5. Numeric variables were robust-scaled for scale-sensitive models.

6. A stratified train-test split was used to preserve class proportions.

This preprocessing design reflects the intended deployment setting: predictions should support future campaign planning rather than explain outcomes after the final call has already occurred.

3 Methods

3.1 Baseline

We used logistic regression as the baseline classifier:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

This model provides a useful reference because it assumes a linear decision structure in the transformed feature space.

3.2 Non-linear Models

To test whether the bank marketing problem contains non-linear predictor-response structure, we retained logistic regression as the main classifier but changed the representation of selected numeric features. We focused on `pdays_clean`, `previous`, and `balance`, because they are both interpretable and directly related to contact history and customer financial status.
The following non-linear variants were implemented:

- **Polynomial Logistic**: quadratic basis expansion for selected numeric predictors.
- **Step Function Logistic**: discretization into bins to capture threshold effects.
- **Spline Logistic**: spline basis functions to capture smooth local non-linearity.
- **GAM-style Logistic**: additive smooth functions for selected predictors:

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + f_1(x_1) + f_2(x_2) + f_3(x_3) + \cdots$$

- **Local Classification (KNN)**: a neighborhood-based classifier used as a local analogue.

3.3 Tree-Based Methods

We evaluated three ensemble tree methods:

- **Bagging**, to reduce variance through bootstrap aggregation,
- **Random Forest**, to decorrelate trees through random feature selection,
- **Gradient Boosting**, to sequentially fit residual error patterns.

These methods are particularly relevant because they model non-linearity and feature interactions without explicit manual basis construction.

3.4 Support Vector Machines

We compared SVM classifiers with linear, polynomial, radial basis function (RBF), and sigmoid kernels. The SVM decision rule takes the form

$$f(x) = \sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b,$$

where $K(\cdot,\cdot)$ is the kernel function. Kernel selection determines whether the induced decision boundary is essentially linear or strongly non-linear. For tabular marketing data, this comparison helps reveal whether the class boundary is globally simple or locally complex.

3.5 Evaluation Metrics

Because the classes are reasonably balanced, we report accuracy, but we also include F1-score and ROC-AUC to avoid over-reliance on a single metric:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \qquad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

ROC-AUC measures ranking quality across decision thresholds, which is useful in campaign targeting where the operating threshold may vary by business objective.

4 Experiments and Results

Table 1 summarizes the main results. The baseline logistic regression achieved an accuracy of 0.6969 and ROC-AUC of 0.7574, providing a solid reference. All transformed logistic models except KNN slightly improved upon this baseline, which suggests that the data contains some non-linear structure. However, the improvements were modest: spline, step-function, GAM-style, and polynomial logistic models remained close to one another. This is a plausible outcome when non-linearity exists but is moderate rather than dominant.

Among the non-linear models, spline logistic regression performed best overall. This suggests that smooth local flexibility is more useful than a single global polynomial expansion. KNN performed slightly worse than the baseline, indicating that purely local neighborhood structure was less informative than a stable global classifier in this feature space. The one-hot encoded, mixed-type structure of the data likely makes nearest-neighbor similarity less reliable.

Tree-based methods delivered the strongest performance. Gradient boosting was the best overall model, followed by random forest. This indicates that interaction effects and hierarchical splits are important in the bank marketing problem, and that manually engineered additive non-linearity is less expressive than ensemble trees for this task.

The SVM results also support the presence of non-linear structure. The RBF kernel outperformed the linear kernel by a clear margin, while the sigmoid kernel performed poorly. Thus, the data is not well separated by a single linear boundary, but it does benefit from a flexible kernel-induced geometry.

Table 1: Comparison of classification models.

Model	Accuracy	F1	ROC-AUC
Gradient Boosting Classifier	0.7367	0.6890	0.7911
Random Forest Classifier	0.7263	0.6830	0.7846
SVM RBF	0.7152	0.6530	0.7687
Bagging Classifier	0.7098	0.6475	0.7686
Spline Logistic	0.7069	0.6510	0.7653
Step Function Logistic	0.7066	0.6543	0.7626
GAM-style Logistic	0.7040	0.6461	0.7627
Polynomial Logistic	0.7026	0.6438	0.7608
Baseline Logistic Regression	0.6969	0.6369	0.7574
Local Classification (KNN)	0.6912	0.6354	0.7516
SVM Linear	0.6643	0.5400	0.7328
SVM Sigmoid	0.5629	0.5297	0.5523

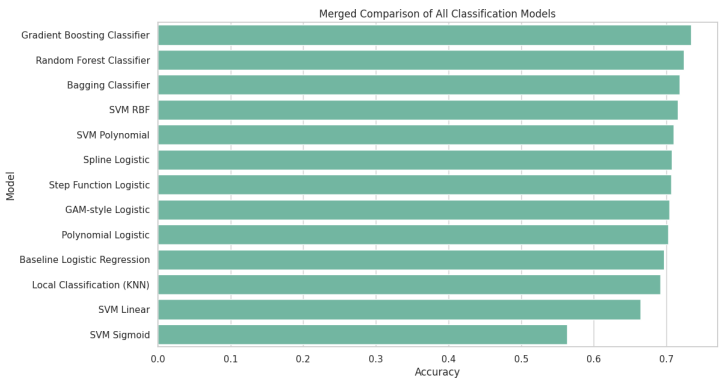

Figure 2: Overall model comparison.

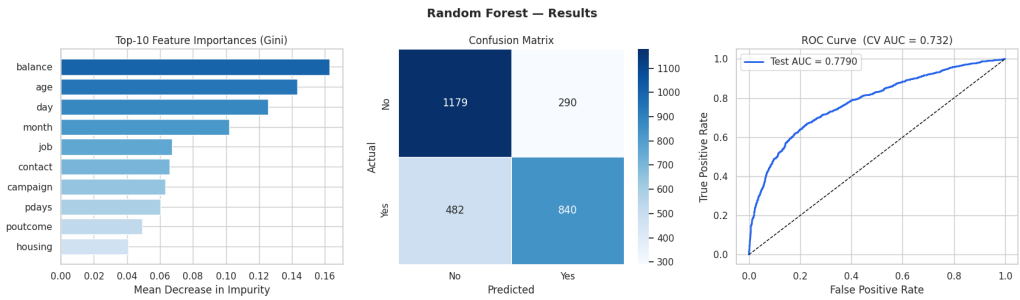

Figure 3: Full diagnostic report for the Random Forest classifier, including classification performance, diagnostic plots, and model interpretation. The figure helps explain the strong overall ranking of Random Forest and highlights its effectiveness in capturing non-linear structure in the bank marketing data.

Permutation importance from the random forest model ranked `month`, `contact`, `poutcome`, `day`, and `housing` as the most influential variables. This suggests that campaign timing, contact channel, and prior campaign outcome are more informative than a purely demographic description of the client.

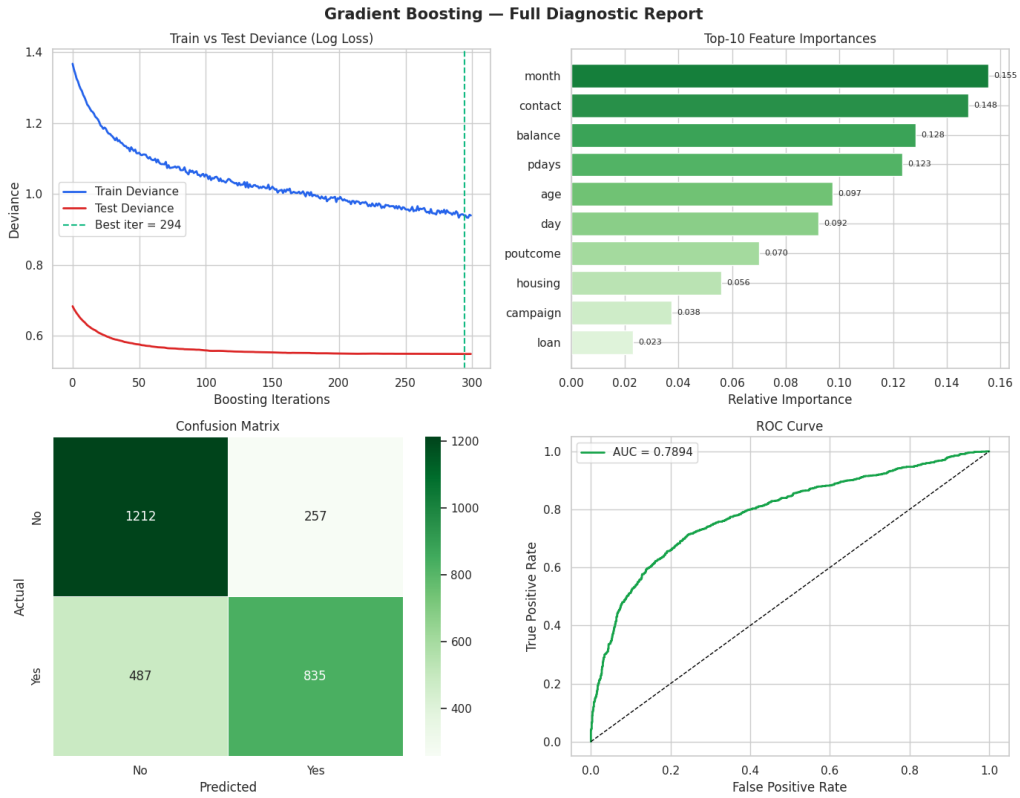

Figure 4: Full diagnostic report for the Gradient Boosting classifier, including performance evaluation and model behavior across the test set.

5 Discussion

The results support three main conclusions. First, the bank marketing problem is not purely linear: non-linear feature expansions consistently improved on the baseline, though only moderately. Second, the strongest gains came from tree ensembles, implying that interaction effects matter at least as much as marginal non-linearities. Third, SVM performance depended strongly

on kernel choice; the superiority of RBF over linear and sigmoid kernels reinforces the view that the decision boundary is non-linear but not arbitrary.
A useful negative result is that KNN and sigmoid SVM underperformed relative to other flexible methods. This indicates that not every non-linear method is equally suitable for structured tabular marketing data. More flexibility is only helpful when it matches the geometry of the problem.

6 Conclusion

We compared baseline logistic regression, non-linear transformed classifiers, tree-based ensembles, and SVM kernels on the Bank Marketing dataset. The best final model was gradient boosting, followed by random forest and RBF-kernel SVM. The results show that the task benefits from flexible decision rules, but that the most effective flexibility comes from models that capture interactions automatically rather than from additive non-linear transformations alone. From a practical perspective, this suggests that future telemarketing campaigns can be improved by data-driven prioritization of clients, especially when campaign history and timing variables are incorporated into the predictive model.

7 Team Contributions

Laman Guluzada conducted exploratory data analysis, designed the preprocessing pipeline, implemented the baseline, non-linear, and SVM models, and contributed to the interpretation and writing. **Cavid Guliyev** implemented the tree-based methods, prepared tree-model visualizations and diagnostics, and contributed to the comparative analysis and writing. Final report preparation, slide design, and result synthesis were completed jointly.

References

[1] S. Moro, P. Cortez, and P. Rita, *A Data-Driven Approach to Predict the Success of Bank Telemarketing*, Decision Support Systems, vol. 62, pp. 22–31, 2014.
Available: `https://www.sciencedirect.com/science/article/abs/pii/S016792361400061X`

[2] J. Bachmann, *Bank Marketing Dataset*, Kaggle.
Available: `https://www.kaggle.com/datasets/janiobachmann/bank-marketing-dataset`

[3] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009.

[4] G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, *An Introduction to Statistical Learning*, 2nd ed., Springer, 2021.

[5] L. Breiman, *Random Forests*, Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[6] J. H. Friedman, *Greedy Function Approximation: A Gradient Boosting Machine*, The Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001.

[7] C. Cortes and V. Vapnik, *Support-Vector Networks*, Machine Learning, vol. 20, pp. 273–297, 1995.